Number theory

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Problem 1 (*Problem 1 IMO 2026*)**.** There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with

$$\gcd(m, n) \quad \text{and} \quad \frac{\operatorname{lcm}(m, n)}{\gcd(m, n)}.$$

He continues to make moves while it is possible to do so.

a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1.

b) Prove that the value of M does not depend on the choices of Confucius.

(Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y, and $\operatorname{lcm}(x, y)$ denotes the least common multiple of x and y.)

Proof. a) Let N be the number of integers strictly greater than 1 currently present on the board. Initially, $N = 2026$.

Consider an arbitrary step where two numbers $m, n > 1$ are chosen. Let $d = \gcd(m, n)$. After this step, the two newly generated numbers on the board are d and $\dfrac{\operatorname{lcm}(m, n)}{d}$.

We define the function $\Omega(x)$ as the total number of prime factors of x (counting multiplicities).

We divide the transformation step into two cases

- In the case with $\gcd(m, n) = 1$, $\operatorname{lcm}(m, n) = mn$. The two new numbers generated are 1 and mn.Since $m, n > 1$, we have $mn > 1$. Thus, replacing two numbers greater than 1 results in only one number greater than 1 being added back. Therefore, the quantity N decreases by 1.
- In the case with $\gcd(m, n) > 1$, consider the sum $\sum \Omega(x)$ over all numbers currently on the board.

  Before the step, the contribution of m and n to $\sum \Omega(x)$ is $\Omega(m) + \Omega(n)$. After the step, the contribution of the two new numbers is $\Omega(\gcd(m, n)) + \Omega\left(\dfrac{\operatorname{lcm}(m, n)}{\gcd(m, n)}\right)$.

Based on the properties of the Ω function, we have

$$\Omega(\gcd(m,n)) + \Omega\left(\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}\right) = \Omega(\operatorname{lcm}(m,n)).$$

On the other hand, since $m \cdot n = \gcd(m,n) \cdot \operatorname{lcm}(m,n)$, it is also true that:

$$\Omega(m) + \Omega(n) = \Omega(\gcd(m,n)) + \Omega(\operatorname{lcm}(m,n))$$

Which implies

$$\Omega(\operatorname{lcm}(m,n)) = \Omega(m) + \Omega(n) - \Omega(\gcd(m,n)).$$

Since $\gcd(m,n) > 1$, we have $\Omega(\gcd(m,n)) > 0$. This leads to

$$\Omega(\operatorname{lcm}(m,n)) < \Omega(m) + \Omega(n).$$

Thus, in this case, N does not increase, but the total sum $\sum \Omega(x)$ of the entire board strictly decreases.

Consider the state defined by the pair of variables $\left(N, \sum \Omega(x)\right)$. Since both N and $\sum \Omega(x)$ are non-negative integers, they cannot decrease indefinitely. Therefore, the transformation process must inevitably terminate after a finite number of steps. Thus, we conclude that the process will be terminated.

b) For each prime p, we consider the quantity

$$E_p = \gcd(v_p(a_1), v_p(a_2), \ldots, v_p(a_{2026})).$$

We will prove that E_p is an invariant throughout the entire process.

Suppose we choose two numbers m, n for a step. Let $\alpha = v_p(m)$ and $\beta = v_p(n)$. The exponents of p in the two new numbers $\gcd(m,n)$ and $\dfrac{\operatorname{lcm}(m,n)}{\gcd(m,n)}$ are respectively

$$v_p(\gcd(m,n)) = \min(\alpha, \beta)$$
$$v_p\left(\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}\right) = \max(\alpha,\beta) - \min(\alpha,\beta) = |\alpha - \beta|.$$

Therefore, looking solely at the exponent of prime p, the operation acts on the exponents as follows

$$(\alpha, \beta) \mapsto (\min(\alpha,\beta), |\alpha - \beta|).$$

Without loss of generality, assume $\alpha \geq \beta$. The transformation then becomes

$$(\alpha, \beta) \mapsto (\beta, \alpha - \beta).$$

By the fundamental property of the Euclidean algorithm, we always have

$$\gcd(\alpha, \beta) = \gcd(\beta, \alpha - \beta).$$

This guarantees that the greatest common divisor of the entire list of exponents v_p on the board never changes after any operation. In other words, the quantity E_p is an invariant for every prime p.

We conclude the problem.

□

Problem 2. Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of integers strictly greater than 1. Suppose that for every positive integer n, the term a_{n+1} is the smallest integer strictly greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for all $i = 1, 2, \ldots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for all positive integers n.

Proof. Let $P(m)$ denote the set of prime divisors of m.

First, the hypothesis implies that for all $i < j$, we have $\gcd(a_i, a_j) > 1$. Indeed, when a_j is chosen, it must share a common divisor strictly greater than 1 with all preceding terms, which includes a_i.

Consequently, the family of sets $\mathcal{F} = \{P(a_n) : n \geq 1\}$ consists of pairwise intersecting finite sets.

Define the set

$$S = \{m \in \mathbb{Z}_{>1} : \gcd(m, a_i) > 1 \text{ for all } i \geq 1\}.$$

We assert that the sequence (a_n) consists precisely of the elements of S that are greater than or equal to a_1, arranged in strictly increasing order.

Clearly, every $a_n \in S$, since any two terms in the sequence share a common divisor greater than 1.

Conversely, assume for the sake of contradiction that there exists $m \in S$ such that $a_n < m < a_{n+1}$. Because $m \in S$, it specifically satisfies $\gcd(m, a_i) > 1$ for all $i = 1, 2, \ldots, n$. However, since $m > a_n$, m would be a valid candidate strictly smaller than a_{n+1} during the selection step for a_{n+1}. This contradicts the minimality of a_{n+1}.Therefore, no element of S can strictly lie between two consecutive terms of the sequence. This implies that (a_n) enumerates exactly the set $S \cap [a_1, \infty)$. □

Problem 3. Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu marks at most n points on the stick, and then Xiang marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu going first. Each player's goal is to maximise the total length of their own pieces.

For each n, determine the largest value c such that Liu may guarantee a total length of at least c, regardless of Xiang's play.

Proof. After the points are marked and the stick is cut into pieces of lengths $x_1, x_2, \ldots, x_k$, the two players take turns choosing any available piece, with Liu going first.

Since each player only cares about maximizing their own total length, the optimal strategy in this phase is: Choose the longest remaining piece on each turn.

Therefore, if we sort the pieces in descending order of length

$$x_1 \geq x_2 \geq \cdots \geq x_k,$$

then Liu will receive a total length of:

$$x_1 + x_3 + x_5 + \ldots$$

In other words, Liu takes the pieces at the odd-indexed positions in the descending sorted list. □

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Problem 4. A function $f : [0, \infty) \to [0, \infty)$ is *max-poly* if there exist finitely many polynomials with real coefficients $p_1(x), p_2(x), \ldots, p_n(x)$ such that

- $p_i(x) \geq 0$ for all $x \geq 0$ and $1 \leq i \leq n$, and
- $f(x) = \max(p_1(x), p_2(x), \ldots, p_n(x))$ for all $x \geq 0$.

Determine whether the function $f(x) = \min(x, x^2)$ is max-poly.

Proof of Problem 4. Suppose, for the sake of contradiction, that f is max-poly. Then, there exists polinomials $p_1, p_2, \cdots, p_n$ s.t $p_i(x) \geq 0$ for all $x \geq 0$ and

$$f(x) = \max\{p_1(x), \cdots, p_r(x)\} \tag{1}$$

for each $x \geq 0$. Obviously, $f(0) = 0$ and thus $p_i(x) = 0$ for each i.

For each $x \in (0, 1)$, one has $f(x) = x^2$. It follows that $0 \leq \dfrac{p_i(x)}{x} \leq x$. Since $p_1, \cdots, p_r$ are polynomials, we have $p_i'(0) = 0$. It implies that $x^2 \mid p_i(x)$ and thus we can write $p_i(x) = x^2 q_i(x)$ ($q_i(x)$ is also a polynomial) for each i.

For each $x \geq 1$, then $f(x) = x$ and we must have $x = x^2 \max\limits_i q_i(x)$. It means that

$$\max_i(x) = \frac{1}{x} \leq 1 \tag{2}$$

or equivalently, c_i is bounded on $[1, +\infty)$. This property yields that $c_i(x) = c_i$ for each i. From (2), we obtain that $c_i = 0$ and hence $0 = \max\limits_i q_i(x) = \dfrac{1}{x}$ for all $x \geq 1$. This is impossible.

Therefore, f is not max-poly. □

Question: Is it true that for any two polynomials p and q, $\min\{p, q\}$ is a max-poly if and only if $p(x) \geq q(x)$ for all $x \geq 0$, or $p(x) \leq q(x)$ for all $x \geq 0$?

Problem 5. Let p be a prime number. Given a square matrix M with entries in $\mathbb{Z}/p\mathbb{Z}$, an *eigenvector* of M is a non-zero vector $\mathbf{v}$ with entries in $\mathbb{Z}/p\mathbb{Z}$ such that $M\mathbf{v} = \lambda\mathbf{v}$ for some $\lambda \in \mathbb{Z}/p\mathbb{Z}$.

For each positive integer n, determine the average number of eigenvectors of an $n \times n$ matrix with entries in $\mathbb{Z}/p\mathbb{Z}$.

(Here $\mathbb{Z}/p\mathbb{Z}$ refers to the integers modulo p.)

Problem 6. Consider the matrix

$$M = \begin{bmatrix} t+1 & t \\ 1 & 1 \end{bmatrix},$$

whose entries are polynomials in the variable t. Let $a(t)$ denote the polynomial in the first row and first column of M^{2025} and let $b(t)$ denote the polynomial in the first row and second column of M^{2025}.

Prove that $a(t)$ has 2025 distinct real roots $\alpha_1, \alpha_2, \ldots, \alpha_{2025}$ and that $b(t)$ has 2025 distinct real roots $\beta_1, \beta_2, \ldots, \beta_{2025}$ satisfying

$$\alpha_1 < \beta_1 < \alpha_2 < \beta_2 < \cdots < \alpha_{2025} < \beta_{2025}.$$

Problem 7. Let V be a finite-dimensional subspace of the real vector space of C^∞ functions from $\mathbb{R}$ to $\mathbb{R}$. Suppose that for all functions $g(x) \in V$, we have $g(2x) \in V$.

Prove that every function in V is a polynomial.

(A function from $\mathbb{R}$ to $\mathbb{R}$ is C^∞ if its k-th derivative exists at every point, for all positive integers k.)

Proof of Problem 5. For each matrix $M \in M_n(\mathbb{Z}_p)$, we define

$$N(M) = \#\left\{v \in \mathbb{Z}_p^n \setminus \{\theta\} : \exists \lambda \in \mathbb{Z}_p, Mv = \lambda v\right\}.$$

Note that $N(M)$ is exactly the number of nonzero eigenvectors of M. To finish, we compute

$$A = \frac{1}{p^{n^2}} \sum_M N(M). \tag{3}$$

Let $\mathcal{S} = \{(M, v) : M \in M_n(\mathbb{Z}_p), v \neq 0, v \text{ is eigenvector of } M\}$. Then $|\mathcal{S}| = \sum_M N(M)$.

Fixing $v \in \mathbb{Z}_p^n, v \neq \theta$. There exist $\lambda \in \mathbb{Z}_p$ such that $Mv = \lambda v$. Since $v \neq 0$, we can choose a suitable basis s.t in this basis, $v = e_1 = (1, 0, \cdots, 0)^T$. It implies that

$$(\lambda, 0, \cdots, 0)^T = \lambda v = Mv = Me_1.$$

There are p ways to choose the first row of M and obviously p^{n^2-n} ways to choose the remaining entries.

Since the number of nonzero vectors is $p^n - 1$, we obtain

$$\mathcal{S} = (p^n - 1)p^{n^2-n+1}.$$

Now, it is easy to have

$$A = p - p^{1-n}$$

□

Proof of Theorem 6. For each positive integer n, we define

$$M^n = \begin{pmatrix} a_n(t) & b_n(t) \\ c_n(t) & d_n(t) \end{pmatrix}.$$

Then $a(t) = a_{2025}(t)$ and $b(t) = b_{2025}(t)$. From $M^{n+1} = M^n M$, we have

$$a_{n+1} = (t+1)a_n + b_n, b_{n+1} = ta_n + b_n. \tag{4}$$

This implies that $a_{n+1} - b_{n+1} = a_n$ and thus

$$a_{n+1} = (t+2)a_n - a_{n-1},$$

and $a_0 = 1$ and $a_1 = t + 1$. Obviously, a_n, b_n are the polynomial of degree n.

We now consider $t \in [-4, 0]$ and let $t = 2\cos\theta - 2$. Then

$$a_{n+1} = 2\cos\theta a_n - a_{n-1}. \tag{5}$$

By induction, we can verify that

$$a_n = \frac{\cos\dfrac{(2n+1)\theta}{2}}{\cos(\theta/2)}. \tag{6}$$

From (6), one has $a_n(t)$ have n roots in $[-4, 0]$ as follow

$$t_j = 2\cos\frac{(2j-1)}{2n+1}\pi - 2$$

for each $j \in \{1, 2\cdots, n\}$. Since $a_n(t)$ is the polynomial of degree n, $t_1, \cdots t_n$ are exactly all roots of $a_n(t)$ on $\mathbb{R}$.

Moreover, by (4), one has

$$b_n(t) = -2\tan\frac{\theta}{2}\sin(n\theta) \tag{7}$$

for all $t \in [-4, 0]$. Similarly, $b_n(t)$ has exactly n roots as follow

$$s_k = 2\cos\frac{k\pi}{n} - 2$$

for each $k = 0, 1, \cdots, n-1$.

In summary, one has

$$\alpha_i = 2\cos\left(1 - \frac{2j}{2n+1}\right)\pi - 2 \tag{8}$$

and

$$\beta_i = 2\cos\frac{n-i}{n}\pi - 2 \tag{9}$$

for each $i = 1, 2, \cdots, n$. Since all angels belong to $[0, \pi]$, we obtain

$$\alpha_1 \le \beta_1 \le \alpha_2 \le \beta_2 \le \cdots \le \alpha_n \le \beta_n$$

for all n.

The result is obtain by applying $n = 2025$. □

Proof of Problem 7 based on Minh Gu'in't. Let $g \in V$ be any arbitrary function. Let $m = \dim V$.Since the vector space V has a dimension of m, the following set of $m+1$ functions

$$g(x), g(2x), g(4x), \ldots, g(2^m x)$$

must be linearly dependent (as they all belong to V).This implies that there exist real constants $a_1, a_2, \ldots, a_m$ such that $g(2^m x) = a_1 g(2^{m-1}x) + \ldots + a_m g(x) \quad \forall x \in \mathbb{R}$.

By setting $f_n(x) = g(2^n x)$, we obtain a sequence of functions $\{f_n(x)\} \subset C^\infty$ that satisfies the linear recurrence relation

$$f_n(x) = a_1 f_{n-1}(x) + \ldots + a_m f_{n-m}(x).$$

According to the theory of linear difference equations (recurrence relations), the general term $f_n(x)$ can be expressed in the form

$$f_n(x) = g(2^n x) = \sum_{j=1}^{\ell}\sum_{i=0}^{b_j-1} C_{i,j}(x)\cdot n^i \cdot r_j^n \quad \forall x \in \mathbb{R}$$

Where $r_1, \ldots, r_\ell$ are the roots (possibly complex numbers) of the characteristic equation associated with the recurrence relation;b_j is the multiplicity of the root r_j, and $C_{i,j}(x)$ are coefficient functions depending on x. Since $g \in C^\infty$, these coefficient functions $C_{i,j}(x)$ also belong to the class C^∞.

Choose a positive integer t large enough such that 2^t is strictly greater than the maximum modulus of all the roots r_j. Specifically, we choose

$$t = \lfloor \log_2(\max_{1 \le j \le \ell} |r_j|) \rfloor + 1$$

This ensures that $|r_j| < 2^t$, which means $\left|\frac{r_j}{2^t}\right| < 1$ for all j. Now, take the t-th derivative of both sides of the general equation with respect to x.

Thus, we have:

$$2^{tn} g^{(t)}(2^n x) = \sum_{j=1}^{\ell} \sum_{i=0}^{b_j - 1} C_{i,j}^{(t)}(x) \cdot n^i \cdot r_j^n.$$

Dividing both sides by 2^{tn} yields

$$g^{(t)}(2^n x) = \sum_{j=1}^{\ell} \sum_{i=0}^{b_j - 1} C_{i,j}^{(t)}(x) \cdot n^i \cdot \left(\frac{r_j}{2^t}\right)^n.$$

Substituting x with $\frac{x}{2^n}$, the equation becomes

$$g^{(t)}(x) = \sum_{j=1}^{\ell} \sum_{i=0}^{b_j - 1} C_{i,j}^{(t)}\left(\frac{x}{2^n}\right) \cdot n^i \cdot \left(\frac{r_j}{2^t}\right)^n \quad \forall x \in \mathbb{R}.$$

Now, we take the limit of both sides as $n \to +\infty$

- Since the functions $C_{i,j}$ are infinitely differentiable, their t-th derivatives are continuous. Therefore, as $n \to +\infty$, $\frac{x}{2^n} \to 0$, which implies $C_{i,j}^{(t)}\left(\frac{x}{2^n}\right) \to C_{i,j}^{(t)}(0)$ (which is a finite constant).
- Since we chose t such that $\left|\frac{r_j}{2^t}\right| < 1$, by the properties of limits, the exponential decay will dominate the polynomial term n^i. That is, $\lim_{n \to +\infty} n^i \cdot \left(\frac{r_j}{2^t}\right)^n = 0$.

Consequently, the entire limit of the Right-Hand Side approaches 0.Therefore, we must have $g^{(t)}(x) = 0 \quad \forall x \in \mathbb{R}$ and, as a result, g is a polynomial. □

Alternative proof for Problem 7. Let $T : V \longrightarrow V, (Tf)(x) \longmapsto f(2x)$. Then T is a linear operator on V. Moreover, T is on-to-one map.

We first prove that if $Tf = \lambda f$, then f is a polynomial. Indeed, we divide into two main cases.

In the case that $\lambda \neq 2^m$ for some m. For each positive integer n, one have $2^n f^{(n)}(0) = \lambda f^{(n)}(0)$ and thus, $f^{(n)}(0) = 0$ for all $n \geq 1$. Fixing $x \in \mathbb{R}$, one has

$$f(x) = \lambda^k f\left(\frac{x}{2^k}\right)$$

for all $k \geq 0$.

By Maclaurin's expression, there exists $\delta_N > 0$ and $C_N > 0$ s.t

$$|f(y)| \leq C_N |y|^{N+1}$$

for all y s.t $|y| \leq \delta_N$ for each $N \geq 0$.

Furthermore, since $f(x) = \lambda^m f\left(\frac{x}{2^m}\right)$, one can choose m such that $\frac{|x|}{2^m} < \delta_N$, and from that we have

$$|f(x)| \leq C_N |x|^{N+1} \left(\frac{|\lambda|}{2^{N+1}}\right)^m.$$

By choosing N such that $\frac{|\lambda|}{2^{N+1}} < 1$ and $m \to \infty$, we obtain that $f(x) = 0$ for all x.

In the case that $\lambda = 2^m$, one can prove that $f^{(k)}(0) = 0$ for all $k \neq m$. Let $c = \frac{f^{(m)}(0)}{m!}$ and define $h(x) = f(x) - cx^m$. Then $h^{(k)}(x) = 0$ for all $k \geq 0$ and

$$h(2x) = 2^m h(2x). \tag{10}$$

Using a similar argument with the previous case, one has $h \equiv 0$ and thus, $f(x) = cx^m$.

By induction, one can prove that for all $r \geq 1$, if $(T - \lambda I)^r f = 0$ then f is a polynomial. Therefore, we obtain the result. □